

代数数论 2026 期末考试

2026 年 6 月 28 日, 时间 10:00–12:00

说明: 满分 100 分, 第一题 40 分, 第二、三题各 30 分, 第四至六题各 20 分。

Problem 1. Let $K = \mathbb{Q}(\sqrt{-15})$ and $\omega := \frac{1+\sqrt{-15}}{2} \in \mathcal{O}_K$.

- (1) Show that $\mathcal{O}_K = \mathbb{Z}[\omega]$ and $\Delta_{K/\mathbb{Q}} = -15$. For a rational prime p , describe how (p) splits in $\mathcal{O}_K$ in terms of the Legendre symbol $\left(\frac{-15}{p}\right)$ (distinguish the cases $p = 2, 3, 5$).
- (2) Let $H := \mathbb{Q}(\sqrt{-15}, \sqrt{-3})$.
 - (a) Prove that $H/\mathbb{Q}$ is abelian. Determine all subextensions of $H/\mathbb{Q}$.
 - (b) Determine all rational primes that ramify in $H/\mathbb{Q}$. And determine the decomposition field and the inertia field of a prime $\mathfrak{p} \subset \mathcal{O}_H$ above those ramified primes.
 - (c) Let p be a prime that is unramified in $H/\mathbb{Q}$. Determine the decomposition field.
- (3) Prove that every ideal class of $\mathcal{O}_K$ contains an ideal of norm at most 2. Show that the ideal $(2, \omega)$ has norm 2 and is not principal. Conclude that the class number $h_K = 2$.
- (4) Deduce that H is the Hilbert class field of K . Show that for every ideal $I \subset \mathcal{O}_K$, the extended ideal $I\mathcal{O}_H$ is principal in $\mathcal{O}_H$. Hint: show that $(2, \omega)\mathcal{O}_H = \beta\mathcal{O}_H$ with $\beta = \frac{\sqrt{5}+\sqrt{-3}}{2} \in H$.

Problem 2. Let p be a prime number.

- (1) Compute explicitly the following p -adic expansions:
 - (a) $\frac{1}{3} = \sum_{n=0}^{\infty} b_n 7^n \in \mathbb{Z}_7$ with $b_n \in \{0, \dots, 6\}, n \leq 6$;
 - (b) $\frac{5}{6} = \sum_{n=n_0}^{\infty} c_n 2^n \in \mathbb{Q}_2$ with $c_n \in \{0, 1\}$ and $c_{n_0} \neq 0$.
- (2) Find the first three non-zero digits of
 - (a) a root of $X^3 - 2X - 1$ in $\mathbb{Z}_3$ that is $\equiv 2 \pmod{3}$.
 - (b) a root of $X^3 - 2$ in $\mathbb{Z}_5$ that is $\equiv 3 \pmod{5}$.
- (3) Let p be a prime. Prove that $x \in \mathbb{Z}_p^\times$ is a square in $\mathbb{Z}_p$ if and only if x is a quadratic residue modulo p when p is odd, and if and only if $x \equiv 1 \pmod{8}$ when $p = 2$.
- (4) Prove that $\mathbb{Z}[[T]]/(T - p) \simeq \mathbb{Z}_p$.

Problem 3. Let K be a number field and let S be a finite set of places of K containing all archimedean places. Set

$$\mathcal{O}_{K,S} := \{x \in K \mid x \in \mathcal{O}_{K_v} \forall \text{ finite } v \notin S\}, \quad U_S := \prod_{v \notin S} \mathcal{O}_{K_v}^\times \times \prod_{v \in S} K_v^\times.$$

Let $\mathcal{I}_S$ be the group of fractional ideals of $\mathcal{O}_K$ coprime to S , and let $\mathcal{P}_S(K) \subset \mathcal{I}_S(K)$ be the subgroup of principal ideals. The S -class group is $\mathcal{Cl}_S(K) := \mathcal{I}_S(K)/\mathcal{P}_S(K)$.

- (1) Prove that $\mathcal{O}_{K,S}$ is a Dedekind domain with $\text{Cart}(\mathcal{O}_{K,S}) \simeq \mathcal{I}_S(K)$. Hint: show that $\mathcal{O}_{K,S}$ is a localization of $\mathcal{O}_K$
- (2) Prove that $\mathbb{A}_K^\times / (K^\times U_S) \simeq \mathcal{Cl}_S(K)$. Show that $\mathcal{Cl}_S(K)$ is a finite group.
- (3) Prove that $\mathbb{A}_K^\times = K^\times U_S$ for some finite set S of places of K .
- (4) Let $U_S^1 = U_S \cap \mathbb{A}_K^{\times,1}$. Show that $\mathcal{O}_{K,S}^\times := K^\times \cap U_S^1$ is the group of units of $\mathcal{O}_{K,S}$ and $U_S^1/\mathcal{O}_{K,S}^\times$ is compact.
- (5) Define a map $\mathcal{L} : U_S \rightarrow \mathbb{R}^S$ by $\mathcal{L}((x_v)_{v \in \Sigma_K}) = (\log |x_v|_v)_{v \in S}$ where $|\cdot|_v$ is the normalized absolute value. Prove that $\mathcal{L}(\mathcal{O}_{K,S}^\times)$ is a lattice in a hyperplane of $\mathbb{R}^S$. Deduce that the group of units $\mathcal{O}_{K,S}^\times$ is a finitely generated abelian group of free rank $|S| - 1$.

Problem 4. Let $p > 3$ be a prime number.

- (1) If $p \equiv 1 \pmod{3}$, show that there exists $u \in \mathbb{Z}_p^\times$ such that u is a non-cube modulo p , and show that $\mathbb{Q}_p(\sqrt[3]{u})$ is the unique unramified cubic extension of $\mathbb{Q}_p$.
- (2) If $p \equiv 1 \pmod{3}$, show that $\mathbb{Q}_p(\sqrt[3]{pu^j})$ for $j = 0, 1, 2$ are the three ramified cubic extensions of $\mathbb{Q}_p$; describe the Galois closure of any one of them.
- (3) If $p \equiv 2 \pmod{3}$, show that $\mathbb{Q}_p(\zeta_{p^2+p+1})$ is the unique unramified cubic extension of $\mathbb{Q}_p$, where ζ_{p^2+p+1} is a primitive $(p^2 + p + 1)$ -th root of unity.
- (4) If $p \equiv 2 \pmod{3}$, show that $\mathbb{Q}_p(\sqrt[3]{p})$ is the unique ramified cubic extension of $\mathbb{Q}_p$; describe the Galois closure of this extension.

Problem 5. Let K be a global function field. Write k for the maximal finite subfield of K (so $|k| = q$ for a prime power q).

- (1) Show that $\mathbb{A}_K^\times / \mathbb{A}_K^{\times,1} \cong \mathbb{Z}$.
- (2) Define $\text{Pic}^0(K) := \text{Div}^0(K) / \text{div}(K^\times)$, where $\text{Div}^0(K) = \{\sum_{v \in \Sigma_K} n_v [v] : \sum_{v \in \Sigma_K} n_v \deg(v) = 0\} \subset \text{Div}(K) = \bigoplus_{v \in \Sigma_K} \mathbb{Z}[v]$. We normalize the Haar measure such that $\text{vol}(\mathcal{O}_{K_v}^\times) = 1$ for each place v . Prove that

$$\text{vol}\left(\mathbb{A}_K^{\times,1}/K^\times\right) = \frac{|\text{Pic}^0(K)|}{q-1}$$

where $K^\times$ is equipped with the counting measure.

Problem 6. Let K be a number field. Call K a CM field if K is totally imaginary and $[K : K^+] = 2$, where K^+ is the maximal totally real subfield of K (equivalently, $K = K^+(\sqrt{d})$ for some totally real K^+ and some $d \in K^+$ all of whose real conjugates are negative).

- (1) Let K be a finite Galois extension of $\mathbb{Q}$. Show that K is either totally real or totally imaginary.
- (2) Let $K \subset \mathbb{C}$ be finite Galois extension of $\mathbb{Q}$. Show that the complex conjugation c lies in the center of $\text{Gal}(K/\mathbb{Q})$ if and only if K is a CM field or K is totally real.
- (3) Show that every subfield $F \subset K$ where K is CM is either totally real or CM.
- (4) Say a number field K has a unit defect if there exists a proper subfield $F \subsetneq K$ such that $\mathcal{O}_K^\times / \mathcal{O}_F^\times$ is finite. Prove that K has a unit defect if and only if K is a CM field.